

INTERNATIONAL SCHOOL

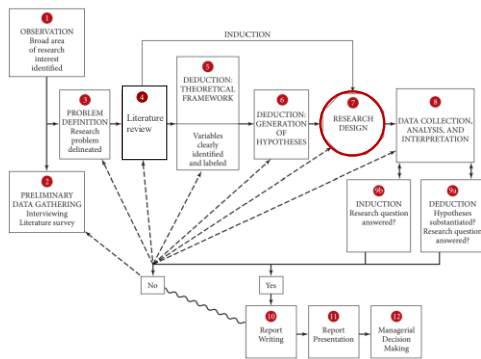
RESEARCH DESIGN

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RESEARCH WORKFLOW



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LEARNING OBJECTIVES

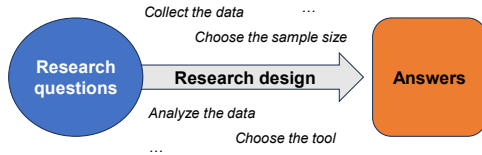
- 1) **Understand** types of research design
- 2) **Choose** a suitable research design for a research topic

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What is research design?

"**Research design** is the **framework** for the collection and analysis of data that will be used to answer a research question or test a hypothesis." (Johnson & Christensen, 2014, p. 32)



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How to Choose a Research Design

Clearly
define
4 main
contents

Research aims or research objectives

Variables

Time

Resources

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Example

Topic: **Mental health status of BDA students of VNUIS and its related factors in 2026**

Objective:

- 1- Describe the mental health status of BDA students of VNUIS in 2024.
- 2- Analyze the related factors of the mental health status of BDA students of VNUIS in 2024.

Key variables: **Mental health status; school year; GPA; family income, join the club; do exercise, etc.**

Time: 2026

Resources: 5 researchers and no money

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Exercise 1

Please write down on paper the name of 01 research topic.

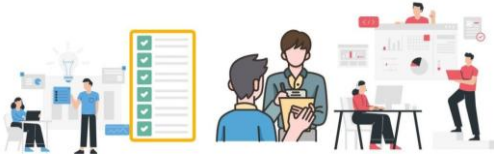
Then, identify the research **objectives** and **key variables** for each research topic.

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Type of research design

Qualitative research Quantitative Research Mix Research



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Qualitative research design

- **Qualitative research methods** are used to collect and analyze non-numerical data.
- Identify, explore, and understand **the nature and cause** of the problem.
- Qualitative research design looks at the **whys and hows**.



Example:

- A researcher wants to study the experience of customers using the service. They conduct in-depth interviews with customers to gather data on their emotional state.

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Quantitative research design

- **Quantitative research methods** are used to collect and analyze **numerical data**.
- **Test a hypothesis**, determine **cause-and-effect relationships**, and measure **the prevalence** of certain phenomena.



Example:

- A company wants to determine the effectiveness of a new advertisement campaign. They survey a large group of people, asking them to rate their awareness of the product and their likelihood of purchasing it.

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Quantitative and Qualitative research design

Qualitative Research	Quantitative Research
Focus on explaining and understanding experiences and perspectives.	Focus on quantifying and measuring phenomena.
Use of non-numerical data, such as words, images, and observations.	Use of numerical data, such as statistics and surveys.
Usually uses small sample sizes.	Usually uses larger sample sizes.
Typically emphasizes in-depth exploration and interpretation.	Typically emphasizes precision and objectivity.
Data analysis involves interpretation and narrative analysis.	Data analysis involves statistical analysis and hypothesis testing.
Results are presented descriptively.	Results are presented numerically and statistically.

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Mix research design

- **Mixed Research design** refers to the combination of **both qualitative and quantitative research methods in a single study**.
- Overcome the limitations of each individual method and to provide a more comprehensive understanding of the research topic.

Example:

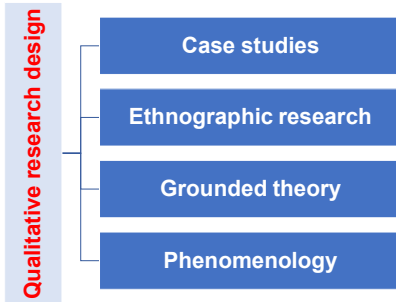
- A university wants to evaluate the effectiveness of a new teaching method in improving student performance. They collect both quantitative data (such as test scores) and qualitative data (such as feedback from students and teachers) to get a complete picture of the impact of the new method.

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Types of qualitative research design



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Case studies

- **Case studies** are a qualitative design in which the researcher explores a program, event, activity, process, or **one or more individuals in depth**.
- The object selected is **rare** or has **very special characteristics**.
- Collecting data through **interviews, observations, and document analysis**.



Example:

- Symptoms and treatment regimen for COVID-19: A case study on the first five patients in Vietnam
- Strategic Turnaround: A Case Study of a Failing Business Revitalized Through Effective Management Practices

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Ethnographic research

- An **ethnographic design** is used to explore the **cultural practices and beliefs** of a group of people.
- This research design involves **living with the group and observing** their daily activities and interactions. The data is then analyzed to identify patterns and themes.



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Ethnographic research



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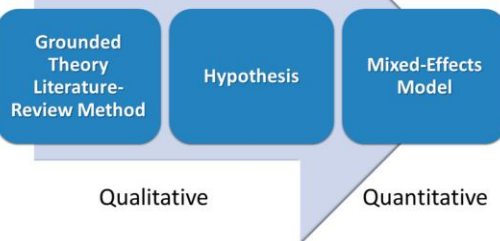
Grounded theory

- **A grounded theory design** is used to **develop a theoretical understanding** of a phenomenon or behavior.
- This type of research design involves collecting data through **interviews** or **observations** and using the data to develop a theory that explains the phenomenon or behavior.
- **Example:** Voices of Women in Leadership: A Grounded Theory Exploration of the Experiences and Challenges.

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Grounded theory

Mixed Methods-
Grounded Theory Design



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Phenomenology design

- A **phenomenological design** is used to explore **the subjective experiences** of individuals.
- This type of research design involves collecting data through **interviews** or **observations** and analyzing the data to identify common themes or patterns.



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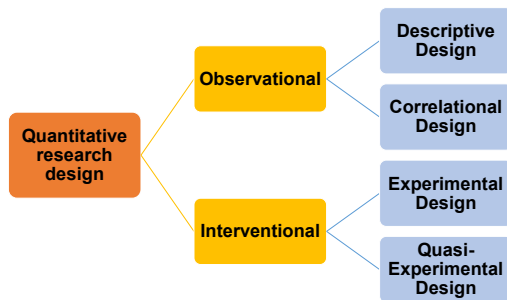
- The Lived Experience of Frontline Healthcare Workers During the COVID-19 Pandemic.
- Embodied Experiences of Virtual Reality Gaming: A Phenomenological Exploration

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Types of quantitative research design



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Types of quantitative research design

Observational design

Observational studies primarily involve the observation of subjects in their natural environment **without any interventions or manipulations** by the researcher.



Interventional design

Intervention studies involve the deliberate **manipulation or intervention by the researcher** to observe the effects of the intervention on the study subjects



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Descriptive Design

- **Descriptive research** is a research method describing the characteristics of the population or phenomenon studied **at a specific time or period of time**.
- The design is described as a photo of the current situation.



- This type of research design involves collecting data through **surveys, observations, and documentary research**.

Example:

- Characterizing online social networking behavior in 2024: A descriptive study of user engagement, posting frequency, and content types.
- Evaluation of job satisfaction levels among employees in company X in 2023: A descriptive study.

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Descriptive Design

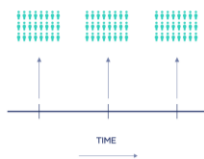
Cross-sectional study

Data collected at one point in time



Longitudinal study

Data collected repeatedly over time



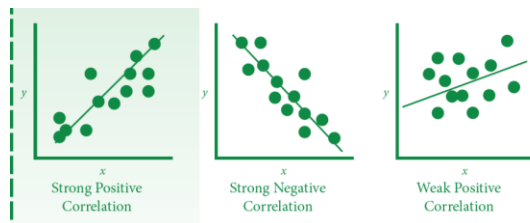
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Correlational Design

- A **correlational design** is used to explore the **relationship between two or more variables without any interventions or manipulations**.
- This type of research design involves collecting data through **surveys, observations, and documentary research**.



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Correlational Design

Example:

- Examining the relationship between **smartphone usage** and **sleep quality** among college students.
- The connection between **physical activity** and **cognitive functioning** in older adults: A cross-sectional study.
- Assessing the relationship between **dietary habits** and **body mass index (BMI)** in a sample of young adults.

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Experimental Design

- An **experimental design** is used to test the **cause-and-effect relationship** between two or more variables through **manipulation or intervention by the researcher**.
- This type of research design involves manipulating one or more independent variables and measuring their effects on a dependent variable.
- Study subjects were divided into **groups randomly**.

Example:

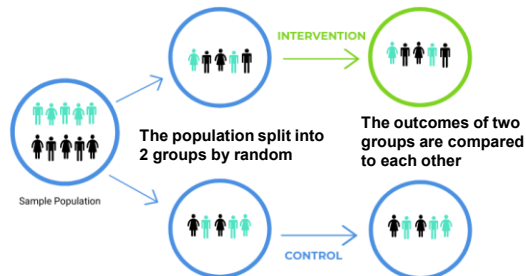
- Examining the efficacy of a **new drug** in lowering blood pressure: **A randomized controlled trial**
- Testing the effectiveness of a **new teaching method** in mathematics: **A randomized controlled trial** in high school classrooms.

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Experimental Design



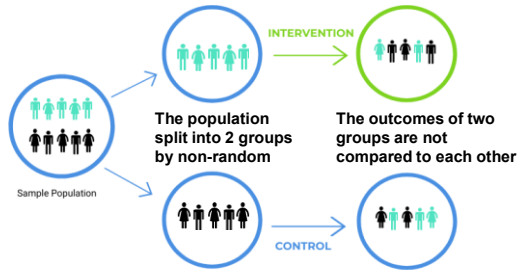
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Quasi-Experimental Design

A **quasi-experimental design** is similar to an experimental design but **lacks the control group or random assignment** that is typical of true experimental designs.

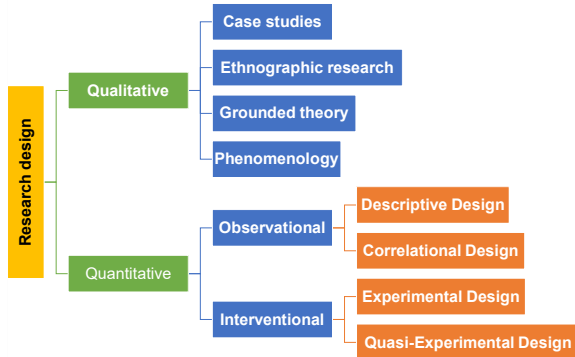


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Summary



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Exercise 2

Predict which type of research design the following research topics belong to, and explain why?

1. Impact of Smartphone Usage on Academic Performance
2. Exploring the Lived Experiences of Single Parents in Urban Settings
3. The Rise and Fall of Nokia: A case study in Corporate Strategy and Technological Disruption
4. Testing the Efficacy of a New Vaccine: A Randomized Clinical Trial
5. Navigating Cultural Identity: An Ethnographic Exploration of an Immigrant Community

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Exercise 3

Please write down on paper the name of 01 research topic.

Then identify which type of research design of that topic, and explain why?

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